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ASSESSMENT TOOL-1 19252505?

1. Identify if the given expressions are balanced or not. i) [9*(k+8-{6+e}+77)/100.

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step-by-step:

[→ opening square bracket

(→ opening parenthesis

{ → opening curly brace

} → closing curly brace

) → closing parenthesis

] → Missing.

ii) (((a*b)+[u/2]-d)

step-by-step:

(→ 3 opening parentheses

[→ opening square bracket

] → closing square bracket

) → only 1 closing parenthesis

Remaining unmatched: ((

2. Given a stack with repeated elements, design as approach to compress it (value & frequency) while mainting stack operations. Apply stack ADT modifications. Analyze trade-off b/w time and space

Example

original stack:

5
5
5
3
3
3
2
2
1

compressed stack:-

(5, 3)
(3, 2)
(2, 3)
(1, 1).

(5, 3) means 5 occur 3 times

(3, 2) means 3 occur 2 times

Modified stack operations:-

Push(x):-

If top value = x → Increase frequency.

Otherwise → Push (x, 1).

POP():-

If frequency > 1 → Decrease frequency.

Otherwise → Remove the top element.

Algorithm:-

```
Push(x)
if stack is empty
    push(x,1).
else if top.value == x.
    top.frequency++
else
    push(x,1).

pop()
if top.frequency > 1
    top.frequency--
else
    pop().
```

Time complexity:-

Operation	complexity
push	O(1)
pop	O(1)
peek	O(1).

Space complexity:-

Normal stack : O(n)

compressed stack: O(K) where K→no.of distinct consective elements

Trade-off b/w Time & space:-

Advantages:-

* Saves memory by storing repeated elements once
* stack operations remains O(1).

Disadvantages

* Each stack node stores an extra frequency field.
* compression work only for consecutive repeated elements.